



DRIVESHIELD

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1. INTRODUCTION & MISSION 🚗

DriveShield is a smart emergency alert system designed to save lives by reducing response times after a crash. By enabling victims to trigger an SOS manually, the system sends real-time alerts to emergency services, ensuring help arrives faster.

2. WHY DRIVESHIELD? ⚠️

Road traffic accidents often result in delayed emergency responses, which significantly impact survival rates. DriveShield aims to bridge this gap by allowing victims to manually trigger an SOS alert after an accident.

3. TECHNOLOGY & FRAMEWORK 🔧

Integrated with ADAS (Advanced Driver Assistance Systems) for crash detection and GPS for precise location tracking. Backend utilizes a cloud database with GSM/LTE communication for reliable cellular transmission in remote areas.

4. SAFETY & PRIVACY MANDATES (20 POINTS) 🚨

1. Zero-Leak Medical Policy: No medical data processed is ever leaked or scraped.
2. End-to-End Encryption: API communications between app and server are fully encrypted.
3. Anti-Tracking: No unauthorized behavioral analytics allowed within the environment.
4. IT Act 2000 Compliance: All operations adhere strictly to Indian cyber laws.
5. APK Integrity: Internal logic to detect and block malicious code modifications.
6. Restricted IAM: Cloud access limited strictly to core system functions via Firebase.
7. Phishing Prevention: Prosecution for creation of "look-alike" websites or APKs.
8. Metadata Protection: No collection of device info without explicit consent.
9. Secure Login: Circumventing VIN-based authentication is a direct violation.
10. Real-Time Tracking Privacy: Location data is only shared upon emergency trigger.
11. Sandbox Isolation: APK data is stored in isolated secure hardware containers.
12. Audit Log Integrity: System logs tracking data access cannot be tampered with.
13. SOS Trigger Validation: Logic to prevent accidental manual alert triggers.
14. Multi-Contact Redundancy: Concurrent alerts to hospital, police, and fire.
15. Secure OTA Updates: Only signed official firmware updates are accepted.
16. Non-Disclosure Agreement: Core team maintains 100% data confidentiality.
17. DPDP Act Alignment: Designed for upcoming data protection guidelines.
18. Anti-Reverse Engineering: Immediate prosecution for decompiling attempts.
19. Remote Area Reliability: Fallback protocols for low-signal environments.
20. Emergency Privacy: System remains dormant until manual SOS is triggered.

5. PLATFORM INTEGRITY

Web Platform: Dashboard for emergency monitoring and response management.

Android Application: Native hardware sensor interface for crash response optimization.

 **INTEGRITY NOTICE: THIS SYSTEM IS A NON-INVASIVE SOFTWARE DEMO. IT DOES NOT TAMPER WITH VEHICLE ELECTRONICS OR VOID OEM WARRANTIES.**

6. LEGAL FRAMEWORK

Designed in alignment with the **Motor Vehicles (Amendment) Act**. Protected under the Indian Copyright Act (Sec 63/63A). Unauthorized replication is strictly prohibited.

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 **OFFICIAL REPOSITORY:** <https://github.com/praneeth-mohan/DriveShield>